

8/3 Forced Vibration of Particles

Although there are many significant applications of free vibrations, the most important class of vibration problems is that where the motion is continuously excited by a disturbing force. The force may be externally applied or may be generated within the system by such means as unbalanced rotating parts. Forced vibrations may also be excited by the motion of the system foundation.

Harmonic Excitation

Various forms of forcing functions $F = F(t)$ and foundation displacements $x_B = x_B(t)$ are depicted in Fig. 8/8. The harmonic force shown in part *a* of the figure occurs frequently in engineering practice, and the understanding of the analysis associated with harmonic forces is a necessary first step in the study of more complex forms. For this reason, we will focus our attention on harmonic excitation.

We first consider the system of Fig. 8/9*a*, where the body is subjected to the external harmonic force $F = F_0 \sin \omega t$, in which F_0 is the force amplitude and ω is the driving frequency (in radians per second). Be sure to distinguish between $\omega_n = \sqrt{k/m}$, which is a property of the system, and ω , which is a property of the force applied to the system. We also note that for a force $F = F_0 \cos \omega t$, one merely substitutes $\cos \omega t$ for $\sin \omega t$ in the results about to be developed.

Courtesy of MTS Systems Corporation



An automobile undergoing vibration testing of its suspension system.

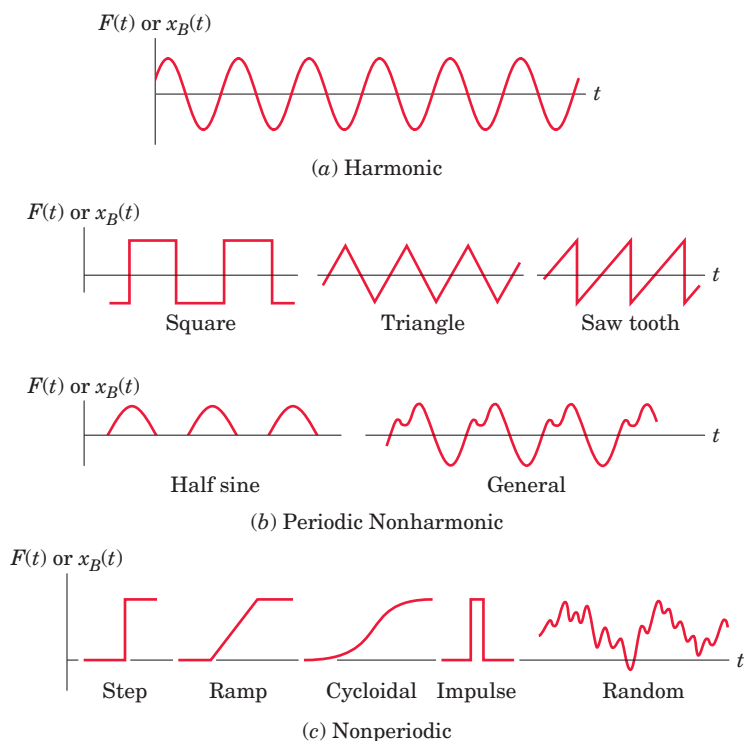


Figure 8/8

From the free-body diagram of Fig. 8/9a, we may apply Newton's second law to obtain

$$-kx - c\dot{x} + F_0 \sin \omega t = m\ddot{x}$$

In standard form, with the same variable substitutions made in Art. 8/2, the equation of motion becomes

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \frac{F_0 \sin \omega t}{m} \quad (8/13)$$

Base Excitation

In many cases, the excitation of the mass is due not to a directly applied force but to the movement of the base or foundation to which the mass is connected by springs or other compliant mountings. Examples of such applications are seismographs, vehicle suspensions, and structures shaken by earthquakes.

Harmonic movement of the base is equivalent to the direct application of a harmonic force. To show this, consider the system of Fig. 8/9b where the spring is attached to the movable base. The free-body diagram shows the mass displaced a distance x from the neutral or equilibrium position it would have if the base were in its neutral position. The base, in turn, is assumed to have a harmonic movement $x_B = b \sin \omega t$. Note that the spring deflection is the difference between the inertial displacements of the mass and the base. From the free-body diagram, Newton's second law gives

$$-k(x - x_B) - c\dot{x} = m\ddot{x}$$

or

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \frac{kb \sin \omega t}{m} \quad (8/14)$$

We see immediately that Eq. 8/14 is exactly the same as our basic equation of motion, Eq. 8/13, in that F_0 is replaced by kb . Consequently, all the results about to be developed apply to either Eq. 8/13 or 8/14.

Undamped Forced Vibration

First, we treat the case where damping is negligible ($c = 0$). Our basic equation of motion, Eq. 8/13, becomes

$$\ddot{x} + \omega_n^2x = \frac{F_0}{m} \sin \omega t \quad (8/15)$$

The complete solution to Eq. 8/15 is the sum of the complementary solution x_c , which is the general solution of Eq. 8/15 with the right side set to zero, and the particular solution x_p , which is *any* solution to the complete equation. Thus, $x = x_c + x_p$. We developed the complementary solution in Art. 8/2. A particular solution is investigated by assuming

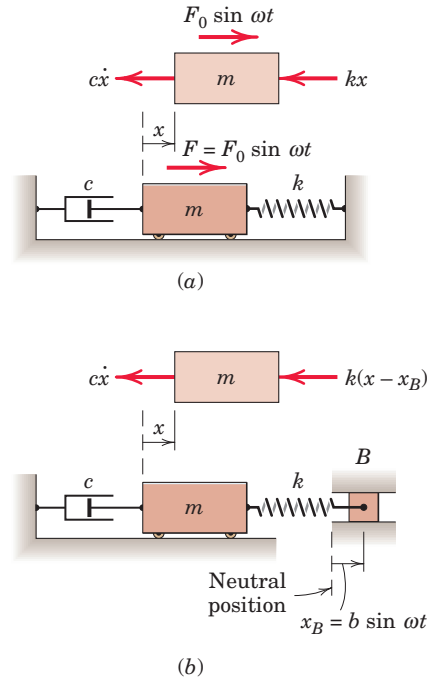


Figure 8/9

that the form of the response to the force should resemble that of the force term. To that end, we assume

$$x_p = X \sin \omega t \quad (8/16)$$

where X is the amplitude (in units of length) of the particular solution. Substituting this expression into Eq. 8/15 and solving for X yield

$$X = \frac{F_0/k}{1 - (\omega/\omega_n)^2} \quad (8/17)$$

Thus, the particular solution becomes

$$x_p = \frac{F_0/k}{1 - (\omega/\omega_n)^2} \sin \omega t \quad (8/18)$$

The complementary solution, known as the *transient solution*, is of no special interest here since, with time, it dies out with the small amount of damping which is always unavoidably present. The particular solution x_p describes the continuing motion and is called the *steady-state solution*. Its period is $\tau = 2\pi/\omega$, the same as that of the forcing function.

Of primary interest is the amplitude X of the motion. If we let δ_{st} stand for the magnitude of the static deflection of the mass under a static load F_0 , then $\delta_{st} = F_0/k$, and we may form the ratio

$$M = \frac{X}{\delta_{st}} = \frac{1}{1 - (\omega/\omega_n)^2} \quad (8/19)$$

The ratio M is called the *amplitude ratio* or *magnification factor* and is a measure of the severity of the vibration. We especially note that M approaches infinity as ω approaches ω_n . Consequently, if the system possesses no damping and is excited by a harmonic force whose frequency ω approaches the natural frequency ω_n of the system, then M , and thus X , increase without limit. Physically, this means that the motion amplitude would reach the limits of the attached spring, which is a condition to be avoided.

The value ω_n is called the *resonant* or *critical frequency* of the system, and the condition of ω being close in value to ω_n with the resulting large displacement amplitude X is called *resonance*. For $\omega < \omega_n$, the magnification factor M is positive, and the vibration is in phase with the force F . For $\omega > \omega_n$, the magnification factor is negative, and the vibration is 180° out of phase with F . Figure 8/10 shows a plot of the absolute value of M as a function of the driving-frequency ratio ω/ω_n .

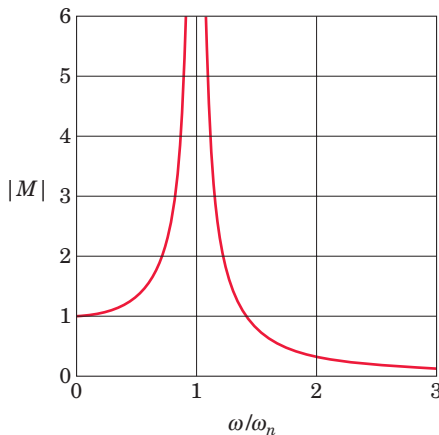


Figure 8/10

Damped Forced Vibration

We now reintroduce damping in our expressions for forced vibration. Our basic differential equation of motion is

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \frac{F_0}{m} \sin \omega t \quad [8/13]$$

Again, the complete solution is the sum of the complementary solution x_c , which is the general solution of Eq. 8/13 with the right side equal to

zero, and the particular solution x_p , which is *any* solution to the complete equation. We have already developed the complementary solution x_c in Art. 8/2. When damping is present, we find that a single sine or cosine term, such as we were able to use for the undamped case, is not sufficiently general for the particular solution. So we try

$$x_p = X_1 \cos \omega t + X_2 \sin \omega t \quad \text{or} \quad x_p = X \sin (\omega t - \phi)$$

Substitute the latter expression into Eq. 8/13, match coefficients of $\sin \omega t$ and $\cos \omega t$, and solve the resulting two equations to obtain

$$X = \frac{F_0/k}{\{[1 - (\omega/\omega_n)^2]^2 + [2\zeta\omega/\omega_n]^2\}^{1/2}} \quad (8/20)$$

$$\phi = \tan^{-1} \left[\frac{2\zeta\omega/\omega_n}{1 - (\omega/\omega_n)^2} \right] \quad (8/21)$$

The complete solution is now known, and for underdamped systems it can be written as

$$x = Ce^{-\zeta\omega_n t} \sin (\omega_d t + \psi) + X \sin (\omega t - \phi) \quad (8/22)$$

Because the first term on the right side diminishes with time, it is known as the *transient solution*. The particular solution x_p is the *steady-state solution* and is the part of the solution in which we are primarily interested. All quantities on the right side of Eq. 8/22 are properties of the system and the applied force, except for C and ψ (which are determinable from initial conditions) and the running time variable t .



KEY CONCEPTS

Magnification Factor and Phase Angle

Near resonance the magnitude X of the steady-state solution is a strong function of the damping ratio ζ and the nondimensional frequency ratio ω/ω_n . It is again convenient to form the nondimensional ratio $M = X/(F_0/k)$, which is called the *amplitude ratio* or *magnification factor*

$$M = \frac{1}{\{[1 - (\omega/\omega_n)^2]^2 + [2\zeta\omega/\omega_n]^2\}^{1/2}} \quad (8/23)$$

An accurate plot of the magnification factor M versus the frequency ratio ω/ω_n for various values of the damping ratio ζ is shown in Fig. 8/11. This figure reveals the most essential information pertinent to the forced vibration of a single-degree-of-freedom system under harmonic excitation. It is clear from the graph that, if a motion amplitude is excessive, two possible remedies would be to (a) increase the damping (to obtain a larger value of ζ) or (b) alter the driving frequency so that ω is farther from the resonant frequency ω_n . The addition of damping is most effective near resonance. Figure 8/11 also shows that, except for $\zeta = 0$, the magnification-factor curves do not actually peak at $\omega/\omega_n = 1$. The peak for any given value of ζ can be calculated by finding the maximum value of M from Eq. 8/23.

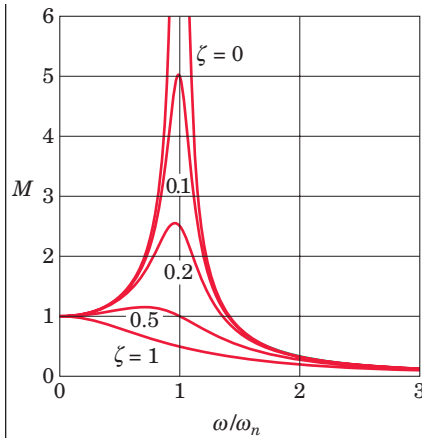


Figure 8/11

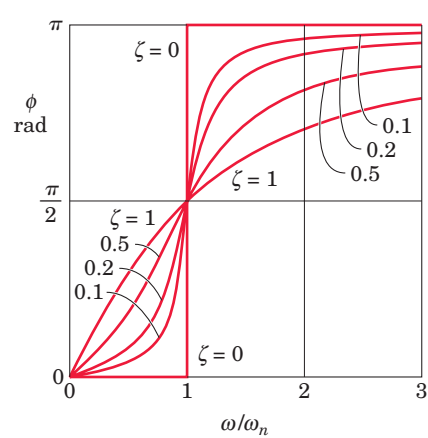


Figure 8/12

The phase angle ϕ , given by Eq. 8/21, can vary from 0 to π and represents the part of a cycle (and thus the time) by which the response x_p lags the forcing function F . Figure 8/12 shows how the phase angle ϕ varies with the frequency ratio for various values of the damping ratio ζ . Note that the value of ϕ when $\omega/\omega_n = 1$ is 90° for all values of ζ . To further illustrate the phase difference between the response and the forcing function, we show in Fig. 8/13 two examples of the variation of F and x_p with ωt . In the first example, $\omega < \omega_n$ and ϕ is taken to be $\pi/4$. In the second example, $\omega > \omega_n$ and ϕ is taken to be $3\pi/4$.

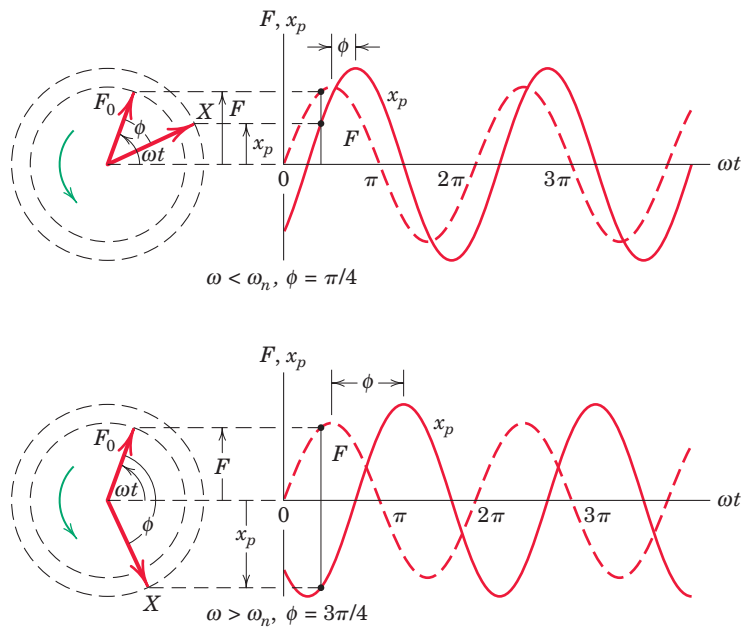


Figure 8/13

Applications

Vibration-measuring instruments such as seismometers and accelerometers are frequently encountered applications of harmonic excitation. The elements of this class of instruments are shown in Fig. 8/14*a*. We note that the entire system is subjected to the motion x_B of the frame. Letting x denote the position of the mass *relative* to the frame, we may apply Newton's second law and obtain

$$-c\dot{x} - kx = m \frac{d^2}{dt^2}(x + x_B) \quad \text{or} \quad \ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = -\ddot{x}_B$$

where $(x + x_B)$ is the inertial displacement of the mass. If $x_B = b \sin \omega t$, then our equation of motion with the usual notation is

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = b\omega^2 \sin \omega t$$

which is the same as Eq. 8/13 if $b\omega^2$ is substituted for F_0/m .

Again, we are interested only in the steady-state solution x_p . Thus, from Eq. 8/20, we have

$$x_p = \frac{b(\omega/\omega_n)^2}{\{[1 - (\omega/\omega_n)^2]^2 + [2\zeta\omega/\omega_n]^2\}^{1/2}} \sin(\omega t - \phi)$$

If X represents the amplitude of the relative response x_p , then the nondimensional ratio X/b is

$$X/b = (\omega/\omega_n)^2 M$$

where M is the magnification ratio of Eq. 8/23. A plot of X/b as a function of the driving-frequency ratio ω/ω_n is shown in Fig. 8/14*b*. The similarities and differences between the magnification ratios of Figs. 8/14*b* and 8/11 should be noted.

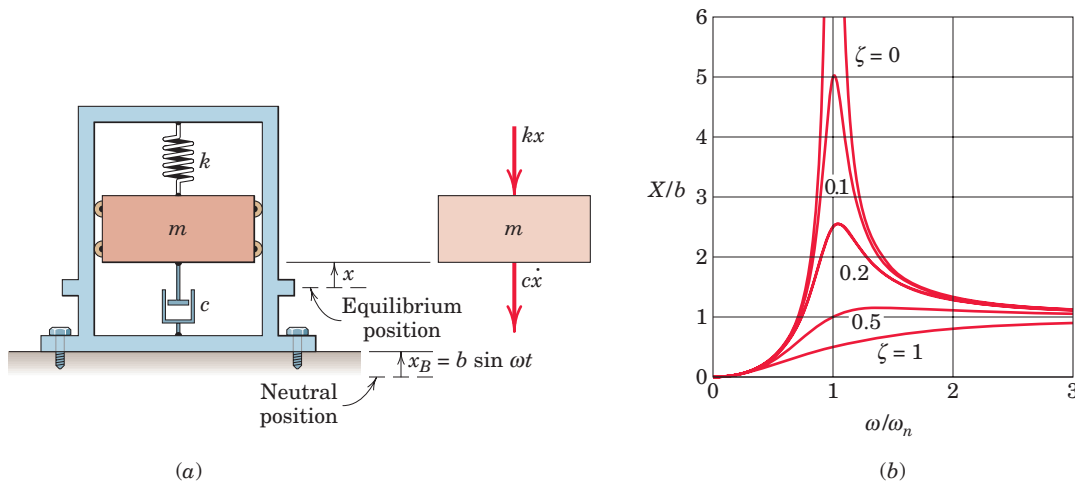


Figure 8/14

If the frequency ratio ω/ω_n is large, then $X/b \cong 1$ for all values of the damping ratio ζ . Under these conditions, the displacement of the mass relative to the frame is approximately the same as the absolute displacement of the frame, and the instrument acts as a *displacement meter*. To obtain a high value of ω/ω_n , we need a small value of $\omega_n = \sqrt{k/m}$, which means a soft spring and a large mass. With such a combination, the mass will tend to stay inertially fixed. Displacement meters generally have very light damping.

On the other hand, if the frequency ratio ω/ω_n is small, then M approaches unity (see Fig. 8/11) and $X/b \cong (\omega/\omega_n)^2$ or $X \cong b(\omega/\omega_n)^2$. But $b\omega^2$ is the maximum acceleration of the frame. Thus, X is proportional to the maximum acceleration of the frame, and the instrument may be used as an *accelerometer*. The damping ratio is generally selected so that M approximates unity over the widest possible range of ω/ω_n . From Fig. 8/11, we see that a damping factor somewhere between $\zeta = 0.5$ and $\zeta = 1$ would meet this criterion.

Electric Circuit Analogy

An important analogy exists between electric circuits and mechanical spring-mass systems. Figure 8/15 shows a series circuit consisting of a voltage E which is a function of time, an inductance L , a capacitance C , and a resistance R . If we denote the charge by the symbol q , the equation which governs the charge is

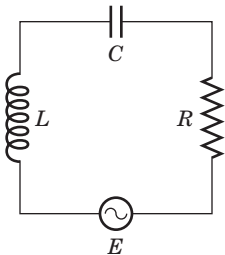


Figure 8/15

$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = E \tag{8/24}$$

This equation has the same form as the equation for the mechanical system. Thus, by a simple interchange of symbols, the behavior of the electrical circuit may be used to predict the behavior of the mechanical system, or vice versa. The mechanical and electrical equivalents in the following table are worth noting:

MECHANICAL-ELECTRICAL EQUIVALENTS

MECHANICAL			ELECTRICAL		
QUANTITY	SYMBOL	SI UNIT	QUANTITY	SYMBOL	SI UNIT
Mass	m	kg	Inductance	L	H henry
Spring stiffness	k	N/m	1/Capacitance	$1/C$	$1/F$ 1/farad
Force	F	N	Voltage	E	V volt
Velocity	$\dot{x}$	m/s	Current	I	A ampere
Displacement	x	m	Charge	q	C coulomb
Viscous damping constant	c	N · s/m	Resistance	R	Ω ohm

SAMPLE PROBLEM 8/4

A 50-kg instrument is supported by four springs, each of stiffness 7500 N/m. If the instrument foundation undergoes harmonic motion given in meters by $x_B = 0.002 \cos 50t$, determine the amplitude of the steady-state motion of the instrument. Damping is negligible.

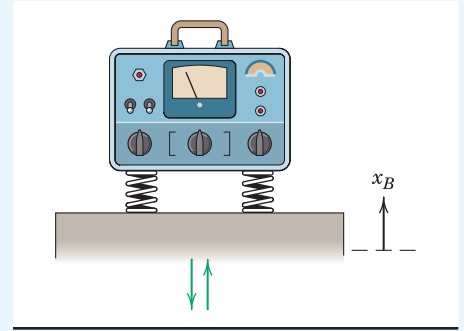
Solution. For harmonic oscillation of the base, we substitute kb for F_0 in our particular-solution results, so that, from Eq. 8/17, the steady-state amplitude becomes

$$1 \quad X = \frac{b}{1 - (\omega/\omega_n)^2}$$

The resonant frequency is $\omega_n = \sqrt{k/m} = \sqrt{4(7500)/50} = 24.5 \text{ rad/s}$, and the impressed frequency $\omega = 50 \text{ rad/s}$ is given. Thus,

$$2 \quad X = \frac{0.002}{1 - (50/24.5)^2} = -6.32(10^{-4}) \text{ m} \quad \text{or} \quad -0.632 \text{ mm} \quad \text{Ans.}$$

Note that the frequency ratio ω/ω_n is approximately 2, so that the condition of resonance is avoided.



Helpful Hints

- 1 Note that either $\sin 50t$ or $\cos 50t$ can be used for the forcing function with this same result.
- 2 The minus sign indicates that the motion is 180° out of phase with the applied excitation.

SAMPLE PROBLEM 8/5

The spring attachment point B is given a horizontal motion $x_B = b \cos \omega t$. Determine the critical driving frequency ω_c for which the oscillations of the mass m tend to become excessively large. Neglect the friction and mass associated with the pulleys. The two springs have the same stiffness k .

Solution. The free-body diagram is drawn for arbitrary positive displacements x and x_B . The motion variable x is measured downward from the position of static equilibrium defined as that which exists when $x_B = 0$. The additional stretch in the upper spring, beyond that which exists at static equilibrium, is $2x - x_B$. Therefore, the *dynamic* spring force in the upper spring, and hence the *dynamic* tension T in the cable, is $k(2x - x_B)$. Summing forces in the x -direction gives

$$[\Sigma F_x = m\ddot{x}] \quad -2k(2x - x_B) - kx = m\ddot{x}$$

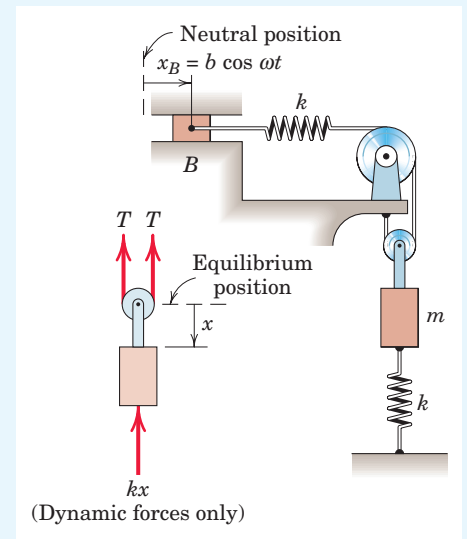
which becomes

$$\ddot{x} + \frac{5k}{m}x = \frac{2kb \cos \omega t}{m}$$

The natural frequency of the system is $\omega_n = \sqrt{5k/m}$. Thus,

$$\omega_c = \omega_n = \sqrt{5k/m}$$

Ans.



Helpful Hints

- 1 If a review of the kinematics of constrained motion is necessary, see Art. 2/9.
- 2 We learned from the discussion in Art. 8/2 that the equal and opposite forces associated with the position of static equilibrium may be omitted from the analysis. Our use of the terms *dynamic* spring force and *dynamic* tension stresses that only the force increments in addition to the static values need be considered.

SAMPLE PROBLEM 8/6

The 100-lb piston is supported by a spring of modulus $k = 200$ lb/in. A dashpot of damping coefficient $c = 85$ lb-sec/ft acts in parallel with the spring. A fluctuating pressure $p = 0.625 \sin 30t$ in lb/in.² acts on the piston, whose top surface area is 80 in.² Determine the steady-state displacement as a function of time and the maximum force transmitted to the base.

Solution. We begin by computing the system natural frequency and damping ratio:

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{(200)(12)}{100/32.2}} = 27.8 \text{ rad/sec}$$

$$\zeta = \frac{c}{2m\omega_n} = \frac{85}{2\left(\frac{100}{32.2}\right)(27.8)} = 0.492 \text{ (underdamped)}$$

The steady-state amplitude, from Eq. 8/20, is

$$X = \frac{F_0/k}{\{[1 - (\omega/\omega_n)^2]^2 + [2\zeta\omega/\omega_n]^2\}^{1/2}}$$

$$= \frac{(0.625)(80)/[(200)(12)]}{\{[1 - (30/27.8)^2]^2 + [2(0.492)(30/27.8)]^2\}^{1/2}}$$

$$= 0.01938 \text{ ft}$$

The phase angle, from Eq. 8/21, is

$$\phi = \tan^{-1} \left[\frac{2\zeta\omega/\omega_n}{1 - (\omega/\omega_n)^2} \right]$$

$$= \tan^{-1} \left[\frac{2(0.492)(30/27.8)}{1 - (30/27.8)^2} \right]$$

$$= 1.724 \text{ rad}$$

The steady-state motion is then given by the second term on the right side of Eq. 8/22:

$$x_p = X \sin(\omega t - \phi) = 0.01938 \sin(30t - 1.724) \text{ ft} \quad \text{Ans.}$$

The force F_{tr} transmitted to the base is the sum of the spring and damper forces, or

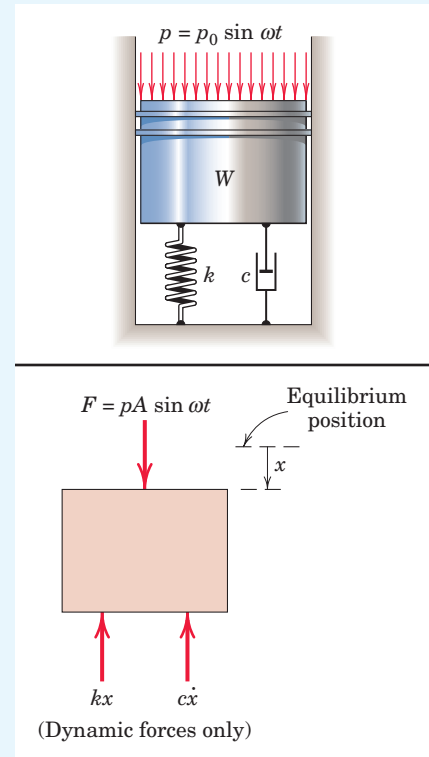
$$F_{tr} = kx_p + c\dot{x}_p = kX \sin(\omega t - \phi) + c\omega X \cos(\omega t - \phi)$$

The maximum value of F_{tr} is

$$(F_{tr})_{\max} = \sqrt{(kX)^2 + (c\omega X)^2} = X\sqrt{k^2 + c^2\omega^2}$$

$$= 0.01938\sqrt{[(200)(12)]^2 + (85)^2(30)^2}$$

$$= 67.9 \text{ lb}$$



Helpful Hints

- 1 You are encouraged to repeat these calculations with the damping coefficient c set to zero so as to observe the influence of the relatively large amount of damping present.
- 2 Note that the argument of the inverse tangent expression for ϕ has a positive numerator and a negative denominator for the case at hand, thus placing ϕ in the second quadrant. Recall that the defined range of ϕ is $0 \leq \phi \leq \pi$.

Ans.